■■ Vector-Shield Streamlit UI Guide & Prototype Handbook

Live Demonstration Breakdown, UI Options Explanation & Future SaaS Evolution

Live Demo: https://vectorshield.streamlit.app/	**Code File:** app/dashboard.py
Lead Architect: Vignesh V S (cb.ai.u4aid23057)	**Milestone:** Review 1 (50% Milestone Complete)

1. Executive Overview: What Prototype Are We Demonstrating NOW?

For Review 1, we demonstrate a **Live Operational Security Cockpit & RAG Middleware Prototype**. The prototype features real-time sentence-transformer embedding (all-MiniLM-L6-v2), sub-2ms Stage 1 LOF interception with Cosine Density Scaling, compiled LangGraph StateGraph self-healing state rewinding, 3D Plotly vector hyperspace scatter plots, and live telemetry audit logging.

2. Detailed Breakdown of Current Streamlit UI Options & Controls

A. Sidebar Controls (app/dashboard.py L49-L74)

• **Stage 1 Safe Ceiling (τ_safe = 1.30):** LOF score cutoff for safe vectors. Scores <= τ_safe are classified as SAFE and passed directly to the LLM.

• **Stage 1 Anomaly Floor (τ_anom = 2.50):** LOF score cutoff for anomalies. Scores >= τ_anom trigger QUARANTINE. The self-healing loop drops the vector and re-retrieves.

• **PCA Density Dimensions (16D):** Number of principal components for dimensionality reduction before LOF scoring.

• **System Status Indicators:** Confirms active sentence-transformers, compiled LangGraph StateGraph, and total benchmark document count.

B. Top KPI Metric Row (app/dashboard.py L80-L89)

• **Stage 1 LOF Latency (1.82 ms):** Measured time taken by Stage 1 to perform PCA reduction and LOF scoring per vector tensor.

• **Clean Pass-Through Rate (96.4%):** Percentage of clean enterprise policy documents (27/28) passed through without false positive quarantine.

• **Attack Detection Rate (100.0%):** Percentage of direct high-dimensional prompt injection outliers (4/4) quarantined by Stage 1.

• **Borderline Routing Ratio (100.0%):** Percentage of stealth injections (3/3) correctly routed to Stage 2 ModernBERT for neural classification.

C. The 4 Dashboard Tabs (app/dashboard.py L93-L251)

• **Tab 1: Live Attack Simulator:** Scenario dropdown + Custom Text Area + Defense Toggle + Execute Button. Demonstrates real-time custom query embedding and LangGraph self-healing logs.

• **Tab 2: 3D Vector Hyperspace:** Interactive Plotly 3D scatter plot showing green clean centroids separated from red outlier attack points.

• **Tab 3: Architecture & Math:** Renders KaTeX LaTeX mathematical formulas (LOF equations, Cosine Scaling) and LangGraph flow diagram.

• **Tab 4: Incident Telemetry Log:** Tabular view of logs/telemetry.json persisting timestamps, LOF scores, latency, and quarantine logs.

3. Future UI Evolution: Phase 2 Enterprise SaaS Security Platform

In Phase 2 (Final Review), Vector-Shield will evolve from a prototype cockpit into a **Production Enterprise Security SaaS Platform** featuring:

• **ModernBERT Attention Heatmap Matrix:** Token-level attention heatmap highlighting exact words inside a document that triggered Stage 2 neural quarantine.

• **HNSW / FAISS Index Manager:** UI controls for managing enterprise vector indexes scaling up to 100,000+ document chunks.

• **Automated Threat Policy Builder:** Rule builder interface allowing security administrators to define custom LOF threshold rules per document category.

• **Multi-Tenant API Portal & SIEM Export:** Dashboard for managing API keys, rate limits, and SIEM webhook integrations (Splunk / Elasticsearch).

• **Real-Time Streaming Log Terminal:** WebSocket-based live log terminal streaming vector scoring events in real-time.